## Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. Calculate the mean of 2, 4, 6.

2. Check if ( $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$) is a distribution.

3. Calculate $(2 - 5)^2$.

4. If $P(A) = 0{,}6$, calculate $P(\overline{A})$.

5. A gross gain of 8 units and a stake of 3 give what net gain?

6. Calculate $\sqrt{0{,}25}$.

**After the diagnostic check.** Go over the flagged concepts with the teacher; redo a similar question, then explain the rule used.

**Oral work and logic.** Explain why a theoretical mean can be an impossible gain. Present the difference between a probability distribution, a simulation and a statistical series.

## Investigate before formalising

Create two games with three equally likely outcomes having expectation 0 but different standard deviations. Check their distributions and explain the difference.

**Organisation.** Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.